

COMMUTE: A Comprehensive Multifaceted Technical Evaluation Framework for Implementation of Auto-Segmentation Models in Radiotherapy

Corresponding Author: Dr Robert Poel

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this manuscript the authors present COMMUTE, a framework for the comparison of autosegmentation methods to be used in segmenting organs at risk during treatment planning. The authors present a method which uses qualitative and quantitative information as part of its data measurement and present the results from an in-house trained 2D U-Net as part of a demonstration of the approach.

COMMUTE as presented would be a novel and useful tool to have in the evolving AI space. AS methods vary in coverage, accuracy at volume, and ability to discern naturally appearing tissue from artefact or lesion. However, the authors present COMMUTE as a framework to compare multiple methods, but only test on a single in-house method. It is difficult to extrapolate an endpoint for COMMUTE as a final discrete quantitative "score" or outcome as the authors do not present one either by contrast with multiple methods or insist on an endpoint. In addition, some tables are not presented correctly comparing two cohorts and are lacking explanation on some choices in data presentation.

The paper spends considerable time engaged in non-scientific discussion. In one example in the Discussion section, the paper describes several different ways across a paragraph the various ways that time can be measured but supports their decision with a few lines at the end of the section. This makes the manuscript difficult to read as the authors conclusions are hidden in other discussion, particularly in the main manuscript. I only found clarity of result and reporting in the appendix where methods are clearly documented without additional comment.

While I believe that the choices of criteria in COMMUTE are useful and well made, the manuscript itself is divided between reporting results of their in-house AS model and describing their framework. To improve the manuscript, the reviewers should aim to acquire other AS methods and compare them using COMMUTE for an intended use-case to demonstrate that COMMUTE is useful to a clinician. For example, evaluating several AS methods and demonstrating that by COMMUTE's scoring criteria, the strongest method determined by COMMUTE yields the highest DSC, Hausdorff distance, least "ABC" between treatment plans, etc.

I would suggest removing the in-house model and leaving COMMUTE to stand alone with other documented methods, so that the manuscript is strengthened by a single conclusion on a single task. This would also remove problematic comparisons in some tables.

I am not a clinician, I did not feel I was able to assess any of the data presented for dose verification.

No AI tools were used in this review, all conclusions are mine and all comments were made by hand.

Reviewer #2

(Remarks to the Author)

Thank you for your submission looking at a validation tool for AS. It is a very novel and timely subject, and this paper is generally very well written

1)Line 75 - repeat of the word but

Reviewer #3

(Remarks to the Author)

First of all congratulation for the work.

Comments:

1. The correction mentioned in the file, please check and adjust it.
2. In-term of language and grammar, writing quality can be improve more and some sentences require revision for better clarity and understanding.
3. Try to avoid personal pronoun throughout the article.
4. Better to add single plan approach in the result and discussion.
5. In dual-plan approach it is better if you fixed one parameter between both plan for a fair comparison.
6. Citation and References style should be adjusted correctly and according to journal guidelines.
7. Figure 8 better to move to method section.
8. Abbreviations in figures and tables should have the full definition in the caption.
9. Abbreviations should be same through all the article.
10. Three cases seem as small number for adjustment evaluation.

Version 1:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

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We would like to thank the reviewers for taking the time to review our submission and their help to improve the manuscript. Hereunder you can find a point by point reply to the comments made by the reviewers. The reviewers text is in **bold** and our response in *italic*. Furthermore we have revised the manuscript and the appendix accordingly of which the result can be seen by the tracked changes function in the word documents of the re-submission.

Reviewer #1 (Remarks to the Author): Attachments

In this manuscript the authors present COMMUTE, a framework for the comparison of autosegmentation methods to be used in segmenting organs at risk during treatment planning. The authors present a method which uses qualitative and quantitative information as part of its data measurement and present the results from an in-house trained 2D U-Net as part of a demonstration of the approach.

First of all, we want to thank the reviewer for the thorough evaluation of the manuscript. This is highly appreciated. We agree with the reviewer that this is indeed a correct understanding of the rationale of the manuscript.

COMMUTE as presented would be a novel and useful tool to have in the evolving AI space. AS methods vary in coverage, accuracy at volume, and ability to discern naturally appearing tissue from artefact or lesion. However, the authors present COMMUTE as a framework to compare multiple methods, but only test on a single in-house method. It is difficult to extrapolate an endpoint for COMMUTE as a final discrete quantitative "score" or outcome as the authors do not present one either by contrast with multiple methods or insist on an endpoint. In addition, some tables are not presented correctly comparing two cohorts and are lacking explanation on some choices in data presentation.

We do think the reviewer is correct here that AS models may behave unpredictably and thus methods for their verification and evaluation are required. Herewith we have to distinguish two aspects. There is an initial evaluation of a newly developed model that will test the general quality and capabilities of this model as a premarket evaluation. Next to that, there is the evaluation and verification of the AS model in its daily use. Both are important and both are topics we are working on, but this specific framework is focused on the former. We do thus not see it directly as a tool but as a framework, or a structure conceptual model that establishes an overarching approach using key principles, recommended methodologies and best practices to guide the analysis. We do not have the intent to provide a tool that will provide a quantitative score for direct comparison. This will require a broader consensus within the field. Quality of AS results can be defined differently and as this framework shows it can reflect different metrics that have their practical and clinical consequences. Especially for dosimetry, there is no consensus on how to draw conclusions what is dosimetrically better. This work is a first step, to inform the community and steer towards possible future guidelines. The expert review is taken in since it is until now a very important factor. We do suggest to add more metrics to validate this evaluation so in the future we could focus more and more on the quantitative parts. We have realized that our ideas are not always well clarified in the manuscript. We also agree that the manuscript, with its hybrid form of framework definition and practical example (partially in the appendix), is unorthodox. However, we think the practical example adds a lot of interesting information and clarification on the framework and is valuable to keep. We have revised the manuscript to clarify the goals of the paper and to guide the reader through the contents on framework and practical example. We hope the reviewer will acknowledge our considerations and is satisfied with the proposed improvements.

The paper spends considerable time engaged in non-scientific discussion. In one example in the Discussion section, the paper describes several different ways across a paragraph the various ways that time can be measured but supports their decision with a few lines at the end of the section. This makes the manuscript difficult to read as the authors conclusions are hidden in other discussion, particularly in the main manuscript. I only found clarity of result and reporting in the appendix where methods are clearly documented without additional comment.

We thank the reviewer for this observation, and have been able to make some improvements here. Although, the main idea of the manuscript is to discuss the different approaches known, point out the strengths and weaknesses and then come to a conclusion of how we would recommend a best practice. This is then exemplified in the extensive methods section in the appendix. Unfortunately, we are not in a position to come to a single conclusion of what is best. This indeed makes the discussion section long, we have tried to make improvements to the comments made by the author and to shorten the discussion section where possible. Also, to summarize the main conclusions we have added the panel containing a summary of the recommendations. The improvements can be found as tracked changes in the revised manuscript.

While I believe that the choices of criteria in COMMUTE are useful and well made, the manuscript itself is divided between reporting results of their in-house AS model and describing their framework. To improve the manuscript, the reviewers should aim to acquire other AS methods and compare them using COMMUTE for an intended use-case to demonstrate that COMMUTE is useful to a clinician. For example, evaluating several AS methods and demonstrating that by COMMUTE's scoring criteria, the strongest method determined by COMMUTE yields the highest DSC, Hausdorff distance, least "ABC" between treatment plans, etc.

We do indeed acknowledge the unorthodox structure of the manuscript. We have tried to improve clarity for the reader to guide them through.

It was however not our goal to make a standardized method or benchmark to directly compare AS methods to each other. We want to provide the community with the knowledge and recommendations to perform more thorough and relevant evaluations in order to improve the interpretation of usefulness and comparability. An actual guideline or benchmark would be an interesting prospect, but we are not there yet.

The report of the in-house model is there to provide as an example, in terms of methods, reporting of results and to show some of the implications of the results, to form as a basis for the discussion to justify the framework. We are not convinced that performing this framework on multiple methods will improve the validation of the framework and might draw too much attention to the comparison of the models and not on the framework. It should also be noted that performing additional international multi-institutional evaluations on additional AS models will require a huge effort and time.

I would suggest removing the in-house model and leaving COMMUTE to stand alone with other documented methods, so that the manuscript is strengthened by a single conclusion on a single task. This would also remove problematic comparisons in some tables.

We understand this recommendations of the reviewer and we as authors have discussed thoroughly on this point as well before submitting. Although, it might improve the clarity of the manuscript, we feel that adding the in-house evaluations provides clear examples and results that strengthen the discussion on the implications of the different methodologies that are reflected in the recommendations.

We did acknowledge your comments on the tables and have tried to address that accordingly to show the audience what is exactly compared to each other. It would have been ideal to have both inter-rater and model results evaluated on the same set of cases. In this case we want to show that it is important to have a benchmark to put the outcome in a usefull perspective. In this case we had inter-rater contouring data on a very similar cohort of patients (same institute, scanners, demographics). This is according to our recommendations a valid comparison. The changes can be found as tracked changes in the revised manuscript.

I am not a clinician, I did not feel I was able to assess any of the data presented for dose verification.

No AI tools were used in this review, all conclusions are mine and all comments were made by hand.

Reviewer #2 (Remarks to the Author):

Thank you for your submission looking at a validation tool for AS. It is a very novel and timely subject, and this paper is generally very well written

1)Line 75 - repeat of the word but

Thanks for catching this error; we have redacted it in the revised manuscript.

Reviewer #3 (Remarks to the Author): Attachments

First of all congratulation for the work.

Comments:

1. The correction mentioned in the file, please check and adjust it.

We want to thank the reviewer for the thorough evaluation of the manuscript. We are thankful for the suggestions, we have redacted these where we found fit in the revised manuscript.

2. In-term of language and grammar, writing quality can be improve more and some sentences require revision for better clarity and understanding.

In terms of language, I have looked at the suggested adjustments. I have tried to see if I could clarify it when necessary. We have improved the overall language in the manuscript by consulting grammaly and have it proofread by a native speaker.

3. Try to avoid personal pronoun throughout the article.

In terms of language, I have looked at the suggested adjustments. I have tried to see if I could clarify it when necessary. We have improved the overall language in the manuscript by consulting grammaly and have it proofread by a native speaker.

4. Better to add single plan approach in the result and discussion.

We do like to clarify that the main reason of bringing up the comparison is showing that there are differences. Both methods can spot dosimetric alterations, but only the two plan approach will show the true direction of the effect and a true comparison of a plan based on AS contours vs one based on GT contours. That is why we defined the two plan approach as the superior and used these as the final method in the evaluation. We have tried to clarify this in the revised manuscript..

5. In dual-plan approach it is better if you fixed one parameter between both plan for a fair comparison.

We are not exactly sure to what is meant with this comment. We think it refers to the method of how to compare the two plans fairly. To clarify, this is what we did: To take into account planning differences, in the two plan approach we left all planning parameters, constraint doses, and weights as well as the beam energies and angles the same. The only thing that was changed were the OAR contours. Finally, we only let the optimizer come up with the optimal dose distribution for the new geometry of the OARs. We have tried to clarify this in the revised appendix.

6. Citation and References style should be adjusted correctly and according to journal guidelines.

I have found the incorrect reference and adjusted it according to the Nature Reference guidelines.

7. Figure 8 better to move to method section.

Since the method section in the main manuscript is very concise, this particular part of the methodology is not discussed in the method section but in the appendix. However, since it is an important issue in the discussion I wanted to add the image there for clarification because the image clarifies the specific issue at hand. I would therefore like to keep it in the discussion section as part of the main manuscript.

The goal of this work was not to compare the methods between dual and single plan. We just try to explain the implications and difference. We have clarified with adding a reference in the appendix for a paper where we used this same methodology.

8. Abbreviations in figures and tables should have the full definition in the caption.

Thanks for pointing this out. I have added the mentioned improvement in the revised manuscript.

9. Abbreviations should be same through all the article.

Thanks for pointing this out. I have added the mentioned improvement in the revised manuscript

10. Three cases seem as small number for adjustment evaluation.

We understand the confusion of the reviewer because the sentence in the manuscript was not clear. To clarify, the adjustment evaluation was performed by 7 experts which evaluated 3 cases each. This meant in total 21 cases containing 17 contours .

If one is referring to the 3 cases, for dosimetric validation of adjusted contours, we want to express that they are there to showcase that contours improved by the physician, will not automatically lead to a favorable dose distribution. It was also performed on 9 cases, but only 3 are shown in the figure. It is not a statement of the quality of the model, it supports our conclusion that the correlation

between contour quality and treatment quality is complex and requires a comprehensive and critical evaluation. We have tried to clarify this in the revised manuscript.